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ADJUSTABLE TIPPING DEVICE FOR A REFUSE VEHICLE

Abstract

A refuse container tipping device includes a first mounting plate, a second mounting plate, a hinge plate, and a plurality of fasteners. The first mounting plate defines a plurality of first slots. The plurality of first slots each extend in a first direction along substantially parallel axes. The second mounting plate is releasably coupled to the first mounting plate and defines a plurality of second slots. The plurality of second slots each extend in a second direction along substantially parallel axes. The first direction is perpendicular to the second direction. The hinge plate is coupled to the second mounting plate. The hinge plate supports a hinge and a grapple arm that is rotatably coupled to the hinge. The plurality of fasteners couple the second mounting plate to the first mounting plate by extending through the plurality of first slots and the plurality of second slots.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/583,380, filed Feb. 21, 2024, which is a continuation of U.S. patent application Ser. No. 17/533,686, filed Nov. 23, 2021, which claims the benefit of and priority to U.S. Provisional Patent Application No. 63/117,541, filed Nov. 24, 2020, each of which are hereby incorporated by reference in their entireties.

BACKGROUND

[0002] Refuse vehicles collect a wide variety of waste, trash, and other material from residences and businesses. Refuse vehicles generally include a lifting system that is movable to engage and lift a waste container so that the waste container's contents can be transferred into a receptacle onboard the refuse vehicle. In rear loading refuse vehicles, the lifting assembly includes a grappling arm that rotates upward and rearward about a hinge on the rear end of the refuse vehicle to lift and tip a waste container off the ground below. The lifting assembly can rotate the waste container so that the opening (e.g., a top of the waste receptacle) faces downward or substantially downward. Gravity urges refuse within the waste container to fall downward, out of the waste container and into the receptacle onboard the refuse vehicle. Once the waste container has been emptied, the lifting assembly can rotate the waste container backward and downward so that the waste container is returned to the ground below, in an upright position.

SUMMARY

[0003] One exemplary embodiment relates to a refuse container tipping device. The refuse container tipping device includes a first mounting plate, a second mounting plate, a hinge plate, and a plurality of fasteners. The first mounting plate defines a plurality of first slots. The plurality of first slots each extend in a first direction along substantially parallel axes. The second mounting plate is releasably coupled to the first mounting plate and defines a plurality of second slots. The plurality of second slots each extend in a second direction along substantially parallel axes. The first direction is perpendicular to the second direction. The hinge plate is coupled to the second mounting plate. The hinge plate supports a hinge and a grapple arm that is rotatably coupled to the hinge. The plurality of fasteners couple the second mounting plate to the first mounting plate. The plurality of fasteners include a bolt and a nut. The bolt extends into and through one of the plurality of first slots and one of the plurality of second slots. The nut is received on the bolt to selectively engage the second mounting plate onto the first mounting plate.

[0004] Another exemplary embodiment relates to a refuse container tipping device. The refuse container tipping device includes a hinge plate that rotatably supports a grapple arm. The hinge plate is coupled to a mounting plate assembly that is configured to be mounted on a tailgate of a refuse vehicle. The mounting plate assembly includes a pair of first mounting plates. The pair of first mounting plates defines at least a plurality of first slots and a plurality of second slots. The plurality of first slots and the plurality of second slots each define paths for slidable motion to adjust the hinge plate relative to the pair of first mounting plates using fasteners that extend through one of the first slots or the second slots.

[0005] Another exemplary embodiment relates to a refuse vehicle. The refuse vehicle includes a chassis supporting a plurality of wheels, a vehicle body supported by the chassis and defining a receptacle for storing refuse, a tailgate positioned at a rear end of the vehicle body and coupled to the receptacle, and a refuse container tipping device coupled to the tailgate. The refuse container tipping device includes a hinge plate that rotatably supports a grapple arm. The hinge plate is coupled to a mounting plate assembly. The mounting plate assembly includes a pair of first mounting plates. The pair of first mounting plates defines at least a plurality of first slots and a plurality of second slots. The plurality of first slots and the plurality of second slots each define paths for slidable motion to adjust the hinge plate relative to the pair of first mounting plates using fasteners that extend through one of the first slots or the second slots.

[0006] The invention is capable of other embodiments and of being carried out in various ways. Alternative exemplary embodiments relate to other features and combinations of features as may be recited herein.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0007] The disclosure will become more fully understood from the following detailed description, taken in conjunction with the accompanying figures, wherein like reference numerals refer to like elements, in which:

[0008] FIG. 1 is a rear perspective view of a refuse vehicle with a tipping device in a first position, according to an exemplary embodiment;

[0009] FIG. 2 is a rear perspective view of the refuse vehicle of FIG. 1, with the tipping device in a second position, according to an exemplary embodiment;

[0010] FIG. 3 is a rear perspective view of a tailgate that can be included on the refuse vehicle of FIG. 1;

[0011] FIG. 4 is a partially exploded view of the tipping device on the tailgate of FIG. 3;

[0012] FIG. 5 is a detailed exploded view of the tipping device of FIG. 4;

[0013] FIG. 6 is a side view of the tailgate of FIG. 3 and the tipping device of FIG. 4;

[0014] FIG. 7 is a top view of the tailgate of FIG. 3, with one of the tipping devices removed to detail a mounting plate assembly within the tipping device of FIG. 4;

[0015] FIG. 8 is a perspective view of a torque plate that can be included within the tipping device of FIG. 4;

[0016] FIG. 9 is a perspective view of a tailgate that can be included on the refuse vehicle of FIG. 1, according to another exemplary embodiment;

[0017] FIG. 10 is a partially exploded view of another mounting plate assembly that can be used to support the tipping device of FIG. 4;

[0018] FIG. 11 is a perspective view of another tailgate that can be included on the refuse vehicle of FIG. 1, according to another exemplary embodiment;

[0019] FIG. 12 is a rear view of the tailgate of FIG. 11, detailing another mounting plate assembly; and

[0020] FIG. 13 is a rear view of another tailgate that can be included on the refuse vehicle of FIG. 1, according to still another embodiment.

DETAILED DESCRIPTION

[0021] Before turning to the figures, which illustrate the exemplary embodiments in detail, it should be understood that the present application is not limited to the details or methodology set forth in the description or illustrated in the figures. It should also be understood that the terminology is for the purpose of description only and should not be regarded as limiting.

[0022] Referring to the FIGURES generally, the various exemplary embodiments disclosed herein

relate to systems, apparatuses, and methods for positioning and operating a waste container tipping device on a refuse vehicle. The waste container tipping device is adjustably mounted to a rear end of the refuse vehicle using mounting plates that define a series of slots. The mounting plates are slidably adjustable relative to one another and relative to the refuse vehicle using fasteners received within the slots. The slots extend in multiple axes (e.g., vertically and horizontally) so that the tipping device can be adjusted upwardly, downwardly, and laterally relative to the refuse vehicle. The adjustable nature of the tipping device allows workers of different heights to customize the refuse vehicle to best accommodate the completion of a refuse route, and further allows for a user to adjust the height of the tipping device to address changes in chassis height that may be caused by a tire size change or suspension lift. The tipping device of the present disclosure is also removable and replaceable, which improves the serviceability of the tipping device relative to conventional tipping devices.

[0023] As shown in FIGS. 1 and 2, a vehicle, shown as refuse truck **10** (e.g., a garbage truck, waste collection truck, sanitation truck, etc.), includes a chassis, shown as a frame **12**, and a body assembly, shown as body **14**, coupled to the frame **12**. The body assembly **14** includes an on-board receptacle **16** and a cab **18**. The cab **18** is coupled to a front end of the frame **12** and includes various components to facilitate operation of the refuse truck **10** by an operator (e.g., a seat, a steering wheel, hydraulic controls, etc.) as well as components that can execute commands automatically to control different subsystems within the vehicle (e.g., computers, controllers, processing units, etc.). In some examples, the cab is configured for autonomous or semi-autonomous driving. The refuse truck **10** further includes a prime mover **20** coupled to the frame **12** at a position beneath the cab **18**. The prime mover **20** provides power to a plurality of motive members, shown as wheels **21**, and to a variety of other systems of the vehicle (e.g., a pneumatic system, a hydraulic system, a power take-off system, etc.). The prime mover **20** may be configured to use one or more of a variety of fuels (e.g., gasoline, diesel, bio-diesel, ethanol, natural gas, etc.), according to various exemplary embodiments. According to an alternative embodiment, the prime mover **20** is one or more electric motors coupled to the frame **12**. The electric motors may consume electrical power from an on-board energy storage device (e.g., batteries, ultra-capacitors, etc.), from an on-board generator (e.g., an internal combustion engine and alternator), from an external power source (e.g., overhead power lines), or a combination of these power supplies to provide power to the systems of the refuse truck **10**.

[0024] According to an exemplary embodiment, the refuse truck **10** is configured to transport refuse from various waste receptacles within a municipality to a storage or processing facility (e.g., a landfill, an incineration facility, a recycling facility, etc.). As shown in FIGS. 1-2, the body **14** and on-board receptacle **16**, in particular, include a series of panels, shown as panels **22**, a cover **24**, and a tailgate **26**. The panels **22**, cover **24**, and tailgate **26** define a collection chamber **28** of the on-board receptacle **16**. Loose refuse is placed into the collection chamber **28**, where it may be thereafter compacted. The collection chamber **28** provides temporary storage for refuse during transport to a waste disposal site or a recycling facility, for example. According to the embodiment shown in FIGS. 1-2, the on-board receptacle **16** and collection chamber **28** are each positioned behind the cab **18**. In the rear-loading refuse truck **10** shown in FIGS. 1-2, loose refuse is emptied into the tailgate **26**. A packer **30** then scoops and pulls the loose refuse forward, out of the tailgate **26** and into the collection chamber **28** for storage and eventual compaction.

[0025] Referring to FIGS. 1-2, the refuse truck **10** is a rear-loading refuse vehicle. In operation, the rear-loading refuse vehicle approaches a pick-up site. With the refuse truck **10** parked, an operator can then position one or more waste containers **100** (often referred to as “toters”) relative to a waste container tipping device **40** that is coupled to the tailgate **26**, at the rear of the refuse truck **10**. The waste container tipping device **40** includes a hydraulically-powered grapple arm **46** that is rotatably coupled to hinges **44** on a hinge plate **42**. The grapple arm **46** includes one or more downward-facing hooks **48** and an upward-facing lip **50** that extend away from the grapple arm **46**. The hooks

48 are arranged to engage a bar extending along a rear of the waste container **100** while the lip **50** engages a handle or ledge **104** formed near a lid of the waste container **100**. By protruding from the grapple arm **46** in opposite directions (e.g., downward-facing hooks **48**, upward-facing lip **50**) to engage different portions of the waste container **100**, the grapple arm **46** is able to securely and releasably couple with the waste container **100** regardless of orientation (e.g., the waste container **100** remains secured throughout the tipping process). In some examples, the downward-facing hooks **48** can slide relative to the grapple arm **46** to more securely engage the bar of the waste container **100**, which in turn causes the lip **50** to more securely engage the handle or ledge **104** of the waste container **100**.

[0026] With the hooks **48** and lip **50** engaged with different components of the waste container **100**, the grapple arm **46** can swing upward and rearward relative to the hinge plate **42** about the hinges **44** to orient an opening **102** of the waste container **100** toward the tailgate **26**. As shown in FIG. 2, the waste container **100** is at least partially inverted so that refuse within the waste container **100** is emptied from the waste container into the on-board receptacle **16**. Aided by gravity and a sudden stop in momentum when the grapple arm **46** reaches its fully inverted position, refuse within the container **100** will fall through the opening **102** and into the tailgate **26** and on-board receptacle **16** below. Accordingly, the waste container tipping device **40** can transition between a fully stowed, or first position (shown in FIG. 1) and a fully deployed, or second position (shown in FIG. 2). Once the contents of the waste container **100** have been emptied, the waste container tipping device **40** rotates back to the first position, and the hooks **48** can translate upward relative to the grapple arm **46** to release the waste container **100** from the waste container tipping device **40**. The grapple arm **46** is configured to rotate at least 90 degrees relative to the hinge plate **42**. In some examples, the grapple arm **46** is configured to rotate at least about 135 degrees relative to the hinge plate **42**.

[0027] The waste container tipping device **40** can be provided with a variety of different motors or power sources to perform the engagement process (e.g., moving the hooks **48** downward) and the tipping process (e.g., rotating the grapple arm **46** about the hinge plate **42** to empty the waste container **100**). In some examples, the grapple arm **46** is hydraulically driven, and provided with pressurized hydraulic fluid from a pump (not shown) onboard the refuse truck **10**. In other examples, a winch system can be used to rotate the grapple arm **46** relative to the tailgate **26**. In still further examples, one or more dedicated electric motors can be provided on the waste container tipping device **40**. The dedicated electric motors can draw power from an alternator or from a battery onboard the refuse truck **10**, for example.

[0028] With additional reference now to FIGS. 3-8, the waste container tipping device **40** is shown with a mounting plate assembly **60** that allows the waste container tipping device **40** to be adjustable in multiple directions relative to the tailgate **26**. The mounting plate assembly **60** includes a series of mounting plates (e.g., mounting plates **62**, **64**, **66**) that define a series of slots (e.g., slots **68**, **70**, **84**, **86**) that allow for sliding motion that can help reposition the hinge plate **42**, hinge **44**, and grapple arm **46** both vertically and laterally relative to the tailgate **26** of the refuse truck **10**. The adjustable nature of the mounting plate assembly **60** can be helpful to accommodate different users or different types of waste containers **100** (e.g., different heights, styles, capacities, etc.). For example, the mounting plate assembly **60** can be readily adjusted based upon the type of route being completed by the refuse truck **10** (e.g., residential v. commercial, waste v. recycling, etc.) so that the waste container tipping device **40** is positioned at a suitable height and location to perform the desired refuse collection process. The adjustable nature of the mounting plate assembly **60** and the waste container tipping device **40**, as a whole, allows the refuse truck **10** to interface with and empty waste containers **100** of different sizes without significant adjustment or labor from a user.

[0029] With specific reference to FIGS. 4-5, the refuse truck **10** is shown with two separate waste container tipping devices **40** mounted to the tailgate **26** or rear fender **52** using mounting plate assemblies **60**. The mounting plate assemblies **60** each include a pair of first mounting plates **62**,

64. The first mounting plates **62, 64** can be directly coupled (e.g., bolted, welded, adhesively mounted) to the tailgate **26** or rear fender **52** of the refuse truck **10**. The first mounting plates **62, 64** are generally defined by a flat mounting face **72** and a flange **74** extending rearwardly and approximately perpendicularly away from the mounting face **72**. The L-shape of the first mounting plates **62, 64** created by the mounting face **72** and the flange **74** forms rectangular nut recesses **78** between the first mounting plates **62, 64** and the fender **52**. As explained in additional detail below, the nut recess **78** creates space for positioning fasteners (e.g., nuts or bolts) relative to the first mounting plates **62, 64** to couple the waste container tipping device **40** to the refuse truck **10**. In some examples, the first mounting plates **62, 64** are symmetrical to one another.

[0030] The first mounting plates **62, 64** define a series of first elongate slots **68, 70**. The first elongate slots **68, 70** are formed entirely through and defined by the mounting face **72**. The first slots **68, 70** have a generally elliptical shape and each extend diagonally (e.g., both horizontally and vertically) across a portion of the mounting face **72**. As depicted in FIG. 5, the first slots **68, 70** are each spaced apart from one another and extend along parallel axes. In some examples, the first slots **68, 70** extend upwardly and outwardly away from the flanges **74** on the first mounting plates **62, 64**. As depicted in FIG. 5, the first slots **68** extend approximately perpendicular to the first slots **70**. In some examples, the mounting face **72** further includes tabs **80** with mounting apertures **82** that can be used in the assembly process, as explained in additional detail below.

[0031] The first mounting plates **62, 64** are configured to removably receive and support a second mounting plate **66**. The second mounting plate **66** has a generally rectangular shape that spans between and can be coupled to each of the first mounting plates **62, 64** in the mounting plate assembly **60** simultaneously. Like the first mounting plates **62, 64**, the second mounting plate **66** includes a plurality of second elongate, elliptically-shaped slots **84, 86** that are defined by and formed through the second mounting plate **66**. The plurality of second slots **84, 86** extend diagonally within the second mounting plate **66** as well. In some examples, the plurality of second slots **84, 86** extend downwardly and outwardly toward a perimeter of the second mounting plate **66**. As depicted in FIG. 5, the second slots **84** extend approximately perpendicular to the first slots **68**, while the second slots **86** extend approximately perpendicular to the first slots **70**. Accordingly, the second slots **84** extend approximately parallel to the first slots **70**, while the second slots **86** extend approximately parallel to the first slots **68**.

[0032] The second mounting plate **66** is removably coupled to the first mounting plates **62, 64** using fasteners, shown as bolts **88** and nuts **90**. The bolts **88** include a widened head **92** and a shank **94**. As depicted in FIGS. 4-5, the shank **94** of each bolt extends through one of the first slots **68, 70** and one of the second slots **84, 86**. The head **92** of each bolt **88** is wider than the second slots **84, 86** so that when the bolts **88** are positioned within the mounting plate assembly **60**, the head **92** of each bolt **88** will engage the second mounting plate **66** and will not pass through the second slots **84, 86**. The shank **94** of each bolt is longer than a combined thickness of the second mounting plate **66** and the mounting face **72** of the first mounting plates **62, 64** so that a portion of each bolt **88** protrudes into the nut recesses **78**. The threaded portions of each shank **94** extending into the nut recesses **78** can then receive one of the nuts **90**. Torqueing the bolts **88** causes the shanks **94** to thread into the nuts **90**, and brings the second mounting plate **66** into engagement with the first mounting plates **62, 64**.

[0033] The shank **94** of each bolt **88** and the orientation of the slots **68, 70, 84, 86** allows the second mounting plate **66** to be both vertically and laterally adjustable relative to the first mounting plates **62, 64**. When the bolts **88** and nuts **90** are not fully tightened, the shank **94** can traverse either of the first slots **68, 70** and second slots **84, 86**, which allows the second mounting plate **66** to move relative to the refuse truck **10**. Because the slots extend in perpendicular directions, the second mounting plate **66** can be adjusted in all four directions (e.g., upward, downward, right, left) relative to the first mounting plates **62, 64**. Once the second mounting plate **66** is positioned at a desired location and orientation relative to the refuse truck **10** and relative to the first mounting

plates **62, 64**, the bolts **88** can be further torqued so that the second mounting plate **66** securely engages the mounting faces **72** of each of the first mounting plates **62, 64** to maintain the mounting plate assembly **60** in position. The length and shape of the slots **68, 70, 84, 86** defines the permissible movement between the second mounting plate **66** and the first mounting plates **62, 64**, such that increasing the length of each slot **68, 70, 84, 86** could increase the degree of adjustability between parts in the adjustable mounting assembly **60**. If further adjustment is needed or if the second mounting plate **66** is preferably removed, the bolts **88** can be loosened (or removed) from the nuts **90**.

[0034] The adjustability of the second mounting plate **66** relative to the first mounting plates **62, 64** allows the entire waste container tipping device **40** to move relative to the refuse truck **10**. As depicted in FIGS. **3-8**, the hinge plate **42** is coupled to the second mounting plate **66**. In some examples, the hinge plate **42** is directly coupled (e.g., welded or bolted) to the rear surface of the second mounting plate **66**, so that movement of the second mounting plate **66** relative to the first mounting plates **62, 64** causes movement of the hinge plate **42**, hinges **44**, and grapple arm **46** relative to the refuse truck **10**. In still other examples, the second mounting plate **66** is also the hinge plate **42**, and directly supports the hinge **44** and grapple arm **46**. The 360 degree adjustability both vertically and laterally relative to the refuse truck **10** allows a worker to easily position the waste container tipping device **40** in a desired and customized location along the refuse truck **10**. The removable coupling formed between the second mounting plate **66** and the first mounting plates **62, 64** further allows the waste container tipping device **40** to be removed for maintenance, replacement, or storage.

[0035] In some examples, the mounting assembly **60** further includes a torque plate **110** to facilitate the coupling process between the bolts **88** and the nuts **90**, which in turn facilitates engagement between the second mounting plate **66** and the first mounting plates **62, 64**. Because there is limited space behind the first mounting plates **62, 64** when the first mounting plates **62, 64** are coupled to the rear fender **52** of the refuse truck **10**, it may be difficult to properly position each of the nuts **90** relative to the first mounting plates **62, 64** before the nuts **90** are threaded to the bolts **88**. The limited spacing between the first mounting plates **62, 64** and the refuse truck **10** can also inhibit tools from being positioned behind the first mounting plates **62, 64**, making it difficult to tighten the nuts **90** to secure the second mounting plate **66** relative to the first mounting plates **62, 64**. The torque plate **110** addresses this issue by providing a series of slots **112** that position and support the nuts **90** in uniformly spaced positions relative to the first mounting plates **62, 64**. The uniformly spaced positions correspond with the distance between adjacent slots **68, 70, 84, 86** formed within the first mounting plates **62, 64** and second mounting plate **66**, so that when the nuts **90** are received within one of the slots **112**, the nuts **90** are also positioned in alignment with one of the first slots **68, 70** and one of the second slots **84, 86**. The torque plate **110** holds the nuts **90** in a stationary position as the shanks **94** of the bolts **88** are threaded into the nuts **90** to form a coupling.

[0036] The slots **112** within the torque plate **110** are designed to receive the nuts **90** and prevent rotation of the nuts **90** to help facilitate the coupling process. As depicted in FIGS. **5** and **8**, the hexagonally-shaped slots **112** (e.g., third slots) extend laterally along the torque plate **110**, and generally include flat upper and lower surfaces **114, 116**. The flat upper and lower surfaces **114, 116** of the slots **112** receive and secure opposite flats formed on the nuts **90**. When the nuts **90** are received within the slots **112**, the flat upper and lower surface **114, 116** interact with the flats formed on the nuts **90** to maintain the orientation of the nuts **90** when the bolts **88** are rotated (e.g., by preventing the nuts **90** from rotating). With rotation of the nuts **90** prevented, rotation of the bolts **88** relative to the nuts **90** will cause the bolts **88** to thread into the nuts **90**. Accordingly, a user can tighten or loosen the coupling between the second mounting plate **66** and the first mounting plates **62, 64** using a single tool (e.g., a socket wrench, nut driver, etc.) rather than needing tools on both the bolt **88** and the nut **90** simultaneously to adjust the coupling.

[0037] The torque plate **110** can move vertically relative to the first mounting plates **62, 64** even

when the torque plate **110** is fastened to the first mounting plates **62, 64**. As depicted in FIGS. **5** and **8**, the torque plate **110** includes locating tabs **118** and locating apertures **120** that are designed to align with the tabs **80** and mounting apertures **82** in the mounting face **72** of the first mounting plates **62, 64**. Using a fastener assembly **122** (e.g., a bolt **124** and nut **126**), torque plates **110** can be threadably coupled to the back side of both of the first mounting plates **62, 64**. With a torque plate **110** coupled to each of the first mounting plates **62, 64**, a nut **90** is aligned with each of the first slots **68, 70**. The locating apertures **120** have an elongate shape which allows vertical adjustment of the torque plate **110** relative to the fastener assembly **122** and relative to the first mounting plates **62, 64**. The elongate shape of the locating apertures allows the torque plate **110** to move relative to the bolt **124**, so that the vertical position of the torque plate **110** relative to the first mounting plates **62, 64** can be adjusted as the second mounting plate **66** is moved. Adjusting the vertical position of the torque plate **110** in turn adjusts the vertical position of the slots **112**, so that the nuts **90** remain in position relative to the torque plate **110** even as the second mounting plate **66** is adjusted. Once the second mounting plate **66** reaches a desired location relative to the first mounting plates **62, 64**, the bolts **88** can be tightened. Because the nuts **90** remain positioned within the hexagonal and elongate slots **112** in the torque plate, the nuts **90** are prevented from rotating. Accordingly, rotation of the bolts **88** will thread the bolts **88** into the nuts **90** to increase the engagement between the first mounting plates **62, 64** and the second mounting plate **66**. The bolts **88** can then be tightened until a secure coupling is formed. To decouple or loosen the second mounting plate **66** for adjustment, the bolts **88** can be rotated in an opposite (e.g., counter-clockwise) direction. As depicted in FIG. **8**, the bottom locating aperture **120** may be formed as a concave recess to maximize the amount of permissible vertical adjustment of the torque plate **110** relative to the first mounting plates **62, 64**.

[0038] Using the adjustable mounting assembly **60** and waste container tipping device **40** described herein, a refuse truck **10** can be readily customized to deal with a variety of different waste receptacles in a way that reduces manual labor needed to empty the waste receptacles. The adjustable nature of the waste container tipping device **40** permits fast assembly and disassembly, which can help promote serviceability and maintenance to the waste container tipping device **40**. Although shown and described in the context of a rear-loading refuse vehicle, the adjustable mounting assembly **60** and waste container tipping device **40** can also be used on front loading or side-loading refuse vehicles. Similarly, the adjustable mounting assembly **60** can be used to adjustably mount a variety of different equipment in customized locations relative to the truck to better help workers perform tasks.

[0039] Various modifications to the waste container tipping device **40** and adjustable mounting assembly **60** can be made using the above-described principles. For example, and as depicted in FIGS. **9-10**, an alternative mounting assembly **160** is provided. The mounting assembly **160** is recessed within the rear fender **52** of the tailgate **26**. The mounting assembly **160** includes a singular first mounting plate **162** that extends along the entire recess within the fender **52**. The first mounting plate **162** includes two sets of slots **168, 170** formed on each side of the mounting plate **162**. The slots **168, 170** extend vertically and upwardly away from the outer perimeter of the mounting plate to operate similarly to the slots **68, 70** described above. A second mounting plate **166** is configured to interact with and move relative to the first mounting plate **162**. The second mounting plate **166** includes opposing elliptically-shaped slots **184, 186**. As described above, the elliptically-shaped slots **168, 170, 184, 186** allow the two mounting plates **162, 166** to move relative to one another to adjust a relative position of the mounting plate **166** relative to the mounting plate **162** and, accordingly, the fender **52**. The two torque plates **110** again provide a series of slots **112** that position and support the nuts **90** in uniformly spaced positions relative to the first mounting plate **162** that help to position and secure the first mounting plate **162** to the second mounting plate **166** by tightening the bolts **88**.

[0040] FIGS. **11-13** depict additional implementations of the present disclosure. For example, adjustable mounting assemblies **260, 360** can be incorporated in a location where the mounting

plates **262**, **362** extend flush with the rear fender **52** of the tailgate **26**. The mounting plates **262**, **362** can be positioned approximately centered within the rear tailgate **26** (as shown in FIG. **11-12**) or positioned offset to the curb side of the rear tailgate **26** (as shown in FIG. **13**). The mounting plates **262**, **362** can be arranged to receive a second mounting plate **66**, **166**, as shown above. The mounting plates **262**, **362** again include elongate slots that allow positional adjustment of the second mounting plate **66**, **166** relative to the mounting plates **262**, **362**. Torque plates **110** with nuts **90** and bolts **88** can once again be used to help secure the second mounting plates **66**, **166** to the mounting plates **262**, **362** in an adjustable fashion, which can move the rear tipping device **40** on the mounting plates **66**, **166** relative to the tailgate **26**. Although consistently described as supporting a rear tipping device, various other equipment can be mounted with the adjustable mounting assemblies **60**, **160**, **260**, **360** described herein. For example, a wash-out pump assembly, different lifting mechanisms, and other accessories can be mounted to the refuse truck **10** using the adjustable mounting assemblies **60**, **160**, **260**, **360**.

[0041] As utilized herein, the terms “approximately”, “about”, “substantially”, and similar terms are intended to have a broad meaning in harmony with the common and accepted usage by those of ordinary skill in the art to which the subject matter of this disclosure pertains. It should be understood by those of skill in the art who review this disclosure that these terms are intended to allow a description of certain features described and claimed without restricting the scope of these features to the precise numerical ranges provided. Accordingly, these terms should be interpreted as indicating that insubstantial or inconsequential modifications or alterations of the subject matter described and claimed are considered to be within the scope of the invention as recited in the appended claims.

[0042] It should be noted that the term “exemplary” as used herein to describe various embodiments is intended to indicate that such embodiments are possible examples, representations, and/or illustrations of possible embodiments (and such term is not intended to connote that such embodiments are necessarily extraordinary or superlative examples).

[0043] The terms “coupled,” “connected,” and the like, as used herein, mean the joining of two members directly or indirectly to one another. Such joining may be stationary (e.g., permanent, etc.) or moveable (e.g., removable, releasable, etc.). Such joining may be achieved with the two members or the two members and any additional intermediate members being integrally formed as a single unitary body with one another or with the two members or the two members and any additional intermediate members being attached to one another.

[0044] References herein to the positions of elements (e.g., “top,” “bottom,” “above,” “below,” “between,” etc.) are merely used to describe the orientation of various elements in the figures. It should be noted that the orientation of various elements may differ according to other exemplary embodiments, and that such variations are intended to be encompassed by the present disclosure.

[0045] It is important to note that the construction and arrangement of the adjustable tipping device as shown in the exemplary embodiments is illustrative only. Although only a few embodiments of the present disclosure have been described in detail, those skilled in the art who review this disclosure will readily appreciate that many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.) without materially departing from the novel teachings and advantages of the subject matter recited. For example, elements shown as integrally formed may be constructed of multiple parts or elements. It should be noted that the elements and/or assemblies of the components described herein may be constructed from any of a wide variety of materials that provide sufficient strength or durability, in any of a wide variety of colors, textures, and combinations. Accordingly, all such modifications are intended to be included within the scope of the present inventions. Other substitutions, modifications, changes, and omissions may be made in the design, operating conditions, and arrangement of the preferred

and other exemplary embodiments without departing from scope of the present disclosure or from the spirit of the appended claims.

Claims

- 1.** A refuse container tipping system comprising: a first mounting plate defining a plurality of first slots, the plurality of first slots each extending along a first plurality of substantially parallel axes; a second mounting plate defining a plurality of second slots, the plurality of second slots each extending along a second plurality of substantially parallel axes, the first plurality of substantially parallel axes being perpendicular to the second plurality of substantially parallel axes; a plurality of fasteners coupling the second mounting plate to the first mounting plate, each of the plurality of fasteners extending into and through one of the plurality of first slots and one of the plurality of second slots to selectively engage the second mounting plate onto the first mounting plate.
- 2.** The refuse container tipping system of claim 1, further comprising a hinge plate coupled to the second mounting plate.
- 3.** The refuse container tipping system of claim 2, the hinge plate supporting a hinge and a grapple arm rotatable coupled to the hinge.
- 4.** The refuse container tipping system of claim 1, wherein the first plurality of substantially parallel axes extend both vertically and horizontally such that the first plurality of substantially parallel axes extend diagonally along the first mounting plate.
- 5.** The refuse container tipping system of claim 4, wherein the first mounting plate is defined by a mounting face and a flange extending perpendicularly away from the mounting face, wherein the plurality of first slots are formed through the mounting face.
- 6.** The refuse container tipping system of claim 5, wherein the second plurality of substantially parallel axes extend both vertically and horizontally such that the second plurality of substantially parallel axes extend diagonally along the second mounting plate.
- 7.** The refuse container tipping system of claim 1, further comprising a torque plate defining a plurality of third slots, the plurality of third slots extending along a third plurality of substantially parallel axes, each of the plurality of fasteners extending into and through one of the plurality of first slots, one of the plurality of second slots, and one of the third plurality of slots.
- 8.** A refuse container tipping system comprising: a hinge plate; and a mounting plate assembly comprising: a plurality of first slots; a plurality of second slots; and a plurality of third slots, the plurality of first slots, the plurality of second slots, and the plurality of third slots each defining paths for slidable motion to adjust a position of at least one of the plurality of first slots, the plurality of second slots, or the plurality of third slots relative to another of the plurality of first slots, the plurality of second slots using fasteners that each extend through the plurality of first slots and one of the plurality of second slots or one of the plurality of third slots; wherein the paths defined by the plurality of first slots are perpendicular to the paths defined by the plurality of second slots.
- 9.** The refuse container tipping system of claim 8, the hinge plate rotatably supporting a grapple arm.
- 10.** The refuse container tipping system of claim 8, wherein the paths defined by the plurality of first slots and the paths defined by the plurality of second slots extend diagonally relative to the paths defined by the plurality of first slots.
- 11.** The refuse container tipping system of claim 8, wherein the plurality of first slots are moveable relative to the plurality of second slots.
- 12.** The refuse container tipping system of claim 8, wherein the plurality of first slots extend in a first direction along substantially parallel axes and the plurality of second slots extend in a second direction along substantially parallel axes.
- 13.** The refuse container tipping system of claim 8, wherein the mounting plate assembly comprises

a pair of first mounting plates each coupled to the hinge plate.

14. The refuse container tipping system of claim 13, wherein the pair of first mounting plates comprises a first plate comprising the plurality of first slots, and a second plate comprising the plurality of second slots.

15. The refuse container tipping system of claim 14, wherein the paths defined by the plurality of first slots extend diagonally along the first plate and the paths defined by the plurality of second slots extend diagonally along the second plate.

16. The refuse container tipping system of claim 13, wherein the mounting plate assembly comprises a pair of torque plates, the pair of torque plates each comprising a portion of the plurality of third slots.

17. The refuse container tipping system of claim 13, wherein the hinge plate is coupled to a second mounting plate assembly, the second mounting plate assembly comprising a plurality of fourth slots, wherein each of the fasteners extend through one of the plurality of fourth slots to couple the hinge plate to the pair of first mounting plates.

18. The refuse container tipping system of claim 13, wherein each of the pair of first mounting plates comprises a mounting face and a flange extending perpendicularly away from the mounting face, wherein the plurality of first slots and the plurality of second slots are formed through the mounting face.

19. The refuse container tipping system of claim 18, wherein each flange of the pair of first mounting plates extends away from the mounting face towards a rear fender of a refuse vehicle and is configured to define a recess between the mounting face and the rear fender, wherein the plurality of third slots are positioned at least partially within one of the recesses.

20. A refuse vehicle comprising: a chassis supporting a plurality of wheels; a vehicle body supported by the chassis and defining a receptacle for storing refuse therein; a tailgate positioned at a rear end of the vehicle body and coupled to the receptacle; and a refuse container tipping device coupled to the tailgate and comprising: an accessory plate; and a mounting plate assembly comprising: a plurality of first slots; a plurality of second slots; and a plurality of third slots, the plurality of first slots, the plurality of second slots, and the plurality of third slots each defining paths for slidable motion to adjust a position of at least one of the plurality of first slots, the plurality of second slots, or the plurality of third slots relative to another of the plurality of first slots, the plurality of second slots using fasteners that each extend through the plurality of first slots and one of the plurality of second slots or one of the plurality of third slots; wherein the paths defined by the plurality of first slots are perpendicular to the paths defined by the plurality of second slots, wherein the plurality of first slots are moveable relative to the plurality of second slots.
